

Bitanu Biswas

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🔗 BitanuCS

Education

Delhi Technological University
M.Tech. by Research in Data Science

Jan 2026 – Present

- GPA: 7.71/10.0

Visva-Bharati University
M.Sc. in Computer Science

Aug 2023 – June 2025

- GPA: 7.49/10.0

University of Calcutta
B.Sc. in Computer Science

July 2018 – June 2021

- GPA: 7.315/10.0

Projects

RAG-Based Regulatory Compliance Chatbot

Jul 2026 - Aug 2026

[GitHub Repo link](#)

[Live Demo](#)

- Built a retrieval-augmented assistant over 27 regulatory documents covering 25 Indian and international frameworks (RBI, SEBI, IRDAI, PMLA, DPDP Act, IT Act, GDPR, ISO 27001), where every claim carries a framework, section, and page citation traceable to the source PDF.
- Enforced grounding with a cosine-distance floor, per-section diversification, and a citation range check, so out-of-corpus questions are refused outright — costing zero LLM calls — instead of being answered from model memory.
- Traced a confidently wrong answer to the ingest layer rather than the prompt, and fixed it structurally: restored lost section boundaries, reflowed column-flattened tables, and re-split the 48.8% of chunks exceeding the embedder's 512-token limit that were being silently truncated.
- Measured retrieval with a 30-case harness requiring no model calls, reporting MRR 0.819, Hit@6 of 92%, and 4/4 correct refusals on out-of-corpus questions.
- **Tools Used:** Python3, LangChain, ChromaDB, BGE embeddings, Sentence-Transformers, HuggingFace Inference API, LangSmith, Streamlit, SQLite, pytest

AI-Powered Job Hunt Assistant

Jun 2026 - Jul 2026

[GitHub Repo link](#)

[Live Demo](#)

- Built an AI tool that reads a resume, infers the roles that fit, and searches across 8 job boards to return a ranked, clutter-free list of matching jobs.
- Designed the workflow in LangGraph, including a human-in-the-loop step that pauses to let the user confirm roles and set filters before searching.
- Ranked results with TF-IDF plus an LLM, surfacing realistically reachable roles above far-off ones.
- **Tools Used:** Python3, LangChain, LangGraph, LangSmith, Streamlit, Pydantic

CNN-based Steganalysis using Modified-AlexNet

Feb 2024 - Jun 2025

[Presentation](#)

[Report](#)

- Developed a modified AlexNet CNN for image steganalysis, adapting the architecture to classify images as cover (clean) versus stego (containing hidden embedded data).
- Augmented the baseline AlexNet with a high-pass noise-residual preprocessing layer and average-pooling stages to amplify the weak steganographic signal, improving sensitivity to subtle pixel-level embedding.

- Trained and evaluated the model on the BOSSbase benchmark (10,000 grayscale images) in Python, achieving 65.25% detection accuracy on the cover/stego classification task.
- **Tools Used:** Python3, TensorFlow, NumPy, CNN, BOSSbase dataset

Signature-Based Steganalysis and Feature-Augmented Password Cracking (FAPC) for Steganography Tool SteganographX Plus

Aug 2024 - Feb 2025

[Presentation](#) 

[Report](#) 

- Built a signature-based steganalysis engine in C to detect BMP images tampered by SteganographX Plus via its byte-level marker.
- Implemented an LSB extraction routine to recover hidden payloads directly from stego BMP pixel data.
- Designed a Feature Augmented Password Cracking (FAPC) module to recover the tool's protection password and unlock concealed data.
- **Tools Used:** C, GCC, GDB, hex editor, SteganographX Plus

Real time Object Detection using OpenCV and YOLOv3

Jan 2021 - Aug 2021

[GitHub Repo link](#) 

[Presentation](#) 

[Report](#) 

- Built a real-time vehicle detection system in Python using OpenCV to identify and classify different vehicle types from video streams.
- Logged each detected vehicle with its type and timestamp, generating a structured record of traffic activity.
- Trained on the COCO dataset to recognize multiple vehicle classes such as cars, trucks, and buses.
- **Tools Used:** Python3, OpenCV, COCO dataset, YOLO

Publications

Signature-Based Steganalysis and Feature-Augmented Password Cracking for Steganography Tool SteganographX Plus

Feb 2026

[Springer Link](#) 

Sayak Bhowal, *Bitanu Biswas*, Madhusudan Paul

DOI: 10.1007/978-3-032-15398-2_12

Experience

Test Analyst

Infosys

Hyderabad, India

Aug 2021 – Jul 2023

- Validated data across 30+ large-scale ETL pipelines, checking millions of records and thousands of business-critical datasets in Informatica PowerCenter and Teradata SQL to ensure 100% data integrity from source to target.
- Worked hands-on with data warehousing structures — Star/Snowflake schemas, metadata, and log tables — verifying loaded data through functional, regression, and progression testing.
- Automated data workflows and historical data loads using Control-M and ESP job schedulers to improve data quality, while mentoring onsite-offshore teams.
- Ran end-to-end test planning, execution, and defect tracking in JIRA and HP ALM across Agile and Waterfall delivery, authoring test plans and reporting to senior stakeholders.

Technical Skills

Languages: Python, C, SQL (Teradata, MySQL)

GenAI, LLMs & Agentic AI: LangGraph, LangChain, LangSmith, RAG, ChromaDB, Sentence-Transformers, HuggingFace, Claude Code

Machine Learning & Deep Learning: TensorFlow, CNNs, OpenCV, NumPy, Pandas

Data Engineering: Informatica PowerCenter, ETL, Data Warehousing, Star/Snowflake schemas

Core CS & Tools: Data Structures & Algorithms (50+ problems solved on [LeetCode](#) ) , Git, GitHub